

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL2- ARTICLE REVIEW

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 10%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Code of Conduct:

I X. Joshua – (192312188) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

X. Joshua

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Task 1: Article Summary

(a) Citation Details and Problem Definition

Khaleghi, T., Murat, A., & Arslanturk, S. (2021). A tree based approach for multi-class classification of surgical procedures using structured and unstructured data. *BMC Medical Informatics and Decision Making*, 21, 328.

The research belongs to the healthcare operations and medical informatics domain. The specific problem is automatically predicting the primary CPT code for a surgical case using information available before surgery. CPT codes are important because they help hospitals plan operating-room schedules, estimate surgery duration, prepare equipment, and support billing. Traditionally, assigning these codes requires considerable manual effort. The authors address this as a multi-class classification problem using a Random Forest model based on decision trees, followed by a proposed Class Weight Recalculation (CWR) method.

(b) Key Objective and Research Gap — 150–200 Words

The main objective of the study is to develop an accurate and practical machine-learning framework for predicting the primary CPT code of a surgical procedure before the operation takes place. The authors identified a research gap in existing surgical CPT prediction research: although machine-learning and deep-learning methods had been used for procedure-code prediction, relatively little work had effectively incorporated the useful information contained in free-text surgical descriptions together with structured hospital data. Manual CPT assignment is also time-consuming and difficult to scale for large hospital databases. To address this gap, the authors proposed a two-stage framework. First, a tuned Random Forest classifier predicts multiple likely CPT codes using structured variables and text-derived features. Second, a novel Class Weight Recalculation method re-ranks the most probable CPT candidates using Random Forest probabilities, feature importance, word frequency, and similarity between medical terms. This combination attempts to exploit both statistical classification and domain-specific textual information. The authors demonstrate that adding text features and the CWR stage can substantially improve CPT prediction accuracy compared with the initial Random Forest model and a neural-network baseline.

Task 2: Methodology Analysis

(a) Machine Learning Technique and Dataset

Machine Learning Technique

The main machine-learning technique is Random Forest (RF), which is an ensemble of decision trees. Random Forest constructs multiple decision trees using bootstrap samples and combines their predictions. For classification, the class receiving the highest number of tree votes is selected. In this study, Random Forest was used for multi-class prediction of surgical CPT codes.

The authors then introduced a second stage called Class Weight Recalculation (CWR). Instead of simply selecting the CPT with the highest Random Forest probability, the method considers the top candidate CPTs and recalculates their weights. The weighting process uses:

- Random Forest prediction probabilities
- Feature-importance values
- Frequency of medical terms
- Similarity between medical terms

- Text-derived TF-IDF information
- Levenshtein distance for word similarity

The re-ranked candidates are then used to improve the final CPT prediction.

Dataset

The study used a real-world historical surgery dataset from a hospital in Southeast Michigan. The original database contained approximately 28,000 scheduled surgery records covering the period from May 2013 to June 2017. After initial data cleaning, approximately 10,000 cases remained for analysis. The data were divided into five specialty groups:

1. Cardiology
2. General surgery
3. Urology
4. Obstetrics and Gynecology (OBGYN)
5. Other specialties

The complete dataset contained 891 unique CPT codes.

The input features included:

- Surgeon ID
- Patient age
- Surgical case type
- Scheduled surgery duration
- TF-IDF-based surgical text features
- Procedure descriptions and medical terminology

The ground-truth CPT labels were obtained from medical coding databases. Because a surgery may have several CPT codes, the authors selected the code with the highest Relative Value Unit (RVU) as the primary CPT code.

Approximately 80% of each specialty dataset was used for training, and the Random Forest hyperparameters were optimized using grid search.

(b) Mapping to CO 4

CO 4 Topic	Relationship to the Article
Inductive Learning	The model learns patterns from previously labeled surgical cases and uses these learned patterns to predict the CPT code of a new surgery.
Decision Trees	Random Forest is an ensemble of multiple decision trees and is the primary classifier used in the study.
Statistical Learning	Random Forest produces class probabilities and feature-importance measures that are used during the classification and CWR stages.

CO 4 Topic	Relationship to the Article
Reinforcement Learning	Not directly used in this research.

Methodological Strength and Limitation

A major strength is the combination of structured and unstructured data. Patient and surgery information alone may not provide enough information to distinguish similar procedures. The authors therefore incorporate information from free-text surgical descriptions using text-mining and TF-IDF features. The CWR stage further uses the information produced by Random Forest to improve the ranking of candidate CPT codes. The authors report that text features improved prediction performance by approximately 20–35%.

One important limitation is the class imbalance and large number of CPT classes. The complete dataset contains more than 800 unique CPT codes, while some codes occur much less frequently than others. This makes it difficult for the classifier to learn reliable patterns for rare procedures. Another limitation is that the dataset comes from a single hospital and is not publicly available, which makes independent reproduction and external validation difficult.

Task 3: Results and Critical Evaluation

(a) Key Results

The authors evaluated the Random Forest and CWR framework using different filtering and evaluation conditions. The important results include:

Model / Method	Accuracy / Performance
Random Forest initial prediction	Approximately 74–76% for primary CPT prediction
Random Forest + CWR	Approximately 8% average improvement over the initial approach
Proposed multi-step approach	Up to 84% accuracy for accuracy@2 under relevant filtering conditions
CWR weighted precision	0.45 on complete data
CWR weighted recall	0.52 on complete data
CWR with F2 filtering – precision	0.62
CWR with F2 filtering – recall	0.64
CWR with F1 + F2 filtering	0.85 precision, 0.86 recall
Neural-network baseline	0.23 precision, 0.22 recall on complete data

The authors also report that incorporating text features increased prediction accuracy by approximately 20–35%. Training time for the framework ranged from approximately 1 to 5 minutes depending on the surgical specialty.

The proposed framework was also compared with a neural-network/LSTM baseline and was reported to achieve better performance in the studied evaluation settings.

(b) Critical Evaluation

The reported results are promising, but they should be interpreted carefully.

First, an accuracy of 84% appears strong, but it is specifically reported for accuracy@2 and under particular filtering conditions. Therefore, it should not be interpreted as saying that the system correctly predicts the exact primary CPT code for 84% of all surgical cases under every condition. Accuracy@2 allows two candidate CPT codes, which naturally provides more opportunity for the correct code to appear in the prediction.

Second, the dataset is highly imbalanced and contains 891 CPT classes. Overall accuracy can therefore hide poor performance on rare CPT codes. Weighted precision and recall partially address this issue, but class-specific metrics such as macro-F1 would provide a better picture of performance across common and rare procedures.

Third, the data originated from a single hospital. Hospital-specific coding practices, surgeon behavior, terminology, and patient populations may influence the learned patterns. Therefore, the model may not generalize equally well to other hospitals.

Fourth, the primary CPT label is derived using the maximum RVU among multiple CPT codes. This is a practical choice, but it may not perfectly represent the clinically most important procedure in every surgery.

Additional Experiments That Would Strengthen the Research

The study could be strengthened by:

1. Testing the framework on datasets from multiple hospitals.
2. Performing external validation using a completely independent healthcare dataset.
3. Reporting macro-F1, micro-F1, balanced accuracy, and per-class recall.
4. Evaluating performance specifically on rare CPT codes.
5. Performing temporal validation by training on earlier years and testing on later years.
6. Comparing CWR with modern NLP approaches such as BERT-based medical language models.
7. Testing SHAP-based feature importance instead of conventional Random Forest feature importance.
8. Measuring the actual improvement in operating-room scheduling and surgery-duration prediction.

The authors themselves suggest using SHAP feature importance and predicting multiple CPT codes as possible future improvements.

(c) Real-World Applicability

The proposed system could be deployed in hospital information systems to assist with:

- Automatic surgical procedure coding
- Operating-room scheduling
- Surgery-duration estimation
- Equipment preparation
- Billing support
- Surgical resource planning
- Reducing manual coding workload

Knowing the probable procedure before surgery can help hospitals allocate operating rooms and equipment more efficiently. The authors specifically identify applications in billing and surgical scheduling.

However, several challenges must be addressed before industry deployment. Healthcare data are sensitive and require strong privacy and security controls. Models trained using one hospital's data may not generalize to other institutions. Medical terminology can also vary between surgeons and hospitals. Furthermore, rare procedures may have insufficient training examples.

Another challenge is explainability. Although Random Forest is generally more interpretable than many deep-learning models, incorrect medical-code predictions could affect billing and scheduling. Therefore, human review should remain part of the workflow, particularly for low-confidence predictions.

Task 4: Reflection and Learning Outcome

(a) Three Key Insights

Insight 1: Inductive Learning Converts Historical Examples into Predictions

The first important insight is that machine learning can learn general patterns from previously labeled examples and apply them to unseen cases. In this article, historical surgical records with known CPT codes are used to learn relationships between patient/surgery characteristics, text features, and procedure codes.

CO 4 connection: This directly demonstrates Inductive Learning, where a model moves from specific training examples to general predictive rules.

Insight 2: Random Forest Demonstrates the Power of Decision-Tree Ensembles

The second insight is that a single decision tree does not always provide the most reliable prediction. Random Forest combines many trees, reducing the dependence on one particular tree and improving classification robustness.

CO 4 connection: This reinforces the Decision Tree topic and demonstrates how ensemble learning can improve upon individual decision trees through multiple-tree voting.

Insight 3: Feature Quality Can Be as Important as the Classifier

The third insight is that improving the input representation can significantly improve machine-learning performance. The researchers found that incorporating text features from surgical descriptions improved prediction accuracy by approximately 20–35%.

CO 4 connection: This demonstrates an important principle of Statistical Learning: model performance depends not only on the selected algorithm but also on how relevant information is represented, selected, and supplied to the learning model.

(b) Proposed Extension

A useful extension would be to develop a multi-hospital CPT prediction system using Random Forest + medical BERT embeddings + SHAP explanations.

The first stage could use the existing structured features such as age, surgeon, surgery type, and scheduled duration. Instead of representing medical text mainly through TF-IDF, a medical language model such as ClinicalBERT or another healthcare-specific transformer could generate contextual representations of the surgical descriptions. These representations could then be combined with structured features and supplied to an ensemble classifier.

SHAP could be used to explain why a particular CPT code was predicted. The model could also be evaluated using data from multiple hospitals, with one hospital completely excluded from training and used only for external testing.

This experiment is feasible because the original study has already demonstrated that combining structured data with text features is beneficial. The proposed extension would test whether modern contextual language representations and explainability techniques can further improve generalization, especially for similar and rare surgical procedures.

Conclusion

This research provides a practical application of Decision Trees, Inductive Learning, and statistical machine learning to healthcare operations. The authors use Random Forest to predict surgical CPT codes and introduce a Class Weight Recalculation method that uses model probabilities, feature importance, and textual information to improve the final prediction.

The study's reported results are encouraging, with the proposed approach reaching up to 84% accuracy@2 under the specified filtering conditions and improving the initial Random Forest results by approximately 8% on average. Text features also provided a substantial improvement.

However, the results should not be interpreted as universally applicable because the dataset came from one hospital, contains significant class imbalance, and the highest accuracy was obtained under specific evaluation conditions.

Overall, the article is highly relevant to CO 4 because it provides a real-world example of how inductive learning and decision-tree ensembles can solve a complex multi-class classification problem.